Deepshik Sharma

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Research Interests: Self-supervised and representation learning; label-efficient learning; model interpretability; foundation models and multimodal learning

EDUCATION

Bachelor of Technology, Computer Science & Engineering — Sep 2021 – Jun 2025
Jain University — Bangalore, India

Capstone Project: *Analyzing Visual Aircraft Representations with Sparse Autoencoders*
[View on arXiv ↗] [View on GitHub ↗]

CGPA: **9/10**

IELTS Academic: **8.5/9** (test taken on Nov 2025)

RESEARCH

PUBLICATIONS

Remote Photoplethysmography: Techniques, Limitations, and Future Directions
Bhardwaj, N., **Sharma, D.**, & Bhavsar, A. (2026). In *Cognitive Biomarkers and BCI Applications* (MBCC 2025), pp. 282–289. Springer, Cham. https://doi.org/10.1007/978-3-032-20634-3_25

PREPRINTS

Analyzing Visual Aircraft Representations with Sparse Autoencoders
Sharma, D. (2026). arXiv:2606.15468. https://doi.org/10.48550/arXiv.2606.15468

An Open-Source End-to-End Pipeline for Large-Scale EEG-Based Brain Age Modelling
Rajesh, S.*, **Sharma, D.***, Venugopal, R., Sasidharan, A., Malipeddi, S., Chowdhury, P., & Ravindra, P. N. (2026). bioRxiv. https://doi.org/10.64898/2026.08.04.742735
* *Equal contribution*

UNDER REVIEW

Self-Supervised ECG Pretraining Under Data and Model Constraints
Sharma, D., Rajesh, S., & Venugopal, R. Submitted to the 16th *IEEE Signal Processing in Medicine and Biology Symposium* (IEEE SPMB 2026), Temple University, Pennsylvania, USA. [View on Google Drive ↗]

CONFERENCE POSTER PRESENTATIONS

Visual Interpretation of Learned Representations in ResNet and Swin Transformer Architectures
Sharma, D., Rajbanshi, R. K., & Sagar, S. Poster, 16th IEEE International Conference on Computing, Communication, and Networking Technologies (ICCCNT 2025), IIT Indore, India, Jul 2025.

ACADEMIC SERVICE & OUTREACH

- **Reviewer** for Computer Vision Track, Springer MBCC 2026 — IIT Mandi, India (Feb 2026)
- **Workshop Facilitator**, "Building Cognitive Experiments" — IIT Mandi, India (Sep 2025)
- **Workshop Organizer**, "Research Methodology and Statistics" — IIT Mandi, India (Mar 2025)

EXPERIENCE

Machine Learning Engineer — July 2026 – Present
NoMind — Thimphu, Bhutan

- R&D Lead for low-resource languages:
  - Building neural machine translation, OCR, and speech capabilities for low-resource Bhutanese languages and dialects (Dzongkha, Tshangla, Kurtoep, Bumthap).
  - Developing language corpora of low-resource dialects for multimodal training data.

Machine Learning Engineer Jan – May 2026
Calyp (formerly **Calypso Brain-Computer Interface**) Bangalore, India

- Developed lightweight and efficient models for passive BCI, micro-gesture and eye-gaze event prediction.
- Reimplemented state-of-the-art deep learning approaches for eye-tracking and BCI on custom datasets.

Research Intern Jul – Nov 2025
National Institute of Mental Health and Neuro Sciences (NIMHANS) Bangalore, India

- Conducted self-supervised pretraining of masked autoencoders for robust representational learning of EEG and ECG signals [See Research/Under Review].
- Developed an open-source pipeline and trained models for the prediction of brain age from EEG data. [See Research/Preprints].
- Conducted research on pupillometry, eye-gaze behavior and visual attention patterns. Developed pipelines for the extraction of eye-tracking features from raw eye-tracker data [View on GitHub ↗].

Research Intern Dec 2024 – Mar 2025
Indian Institute of Technology, Mandi Himachal Pradesh, India

- Worked on deep learning approaches for remote photoplethysmography (rPPG) signal prediction from facial videos [See Research/Publications].
  - Trained and evaluated multiple deep learning models for rPPG signal prediction.
  - Integrated attention modules into 3D-CNN models to examine their effect on predictive performance.

Research Assistant Sep 2024 – Feb 2025
Jain University Bangalore, India

- Integrated fine-tuned large language models (LLMs) into software-defined networking (SDN) paradigms:
  - Automated the generation of natural language sentence-pairs from tabular for LLM fine-tuning.
  - Fine-tuned and evaluated open-source LLMs (T5, LLaMA) to predict network performance.
  - Created visualizations to support research communication, including analysis plots and diagrams.

Applied Machine Learning Intern Jun – Aug 2024
Druk Holding and Investments Thimphu, Bhutan

- Worked on the Automatic Number Plate Recognition (ANPR) System in collaboration with the Royal Bhutan Police.
  - Trained and evaluated the performance of multiple object detection and OCR models.
  - Built a post-processing pipeline to clean and validate OCR predictions. This ensured conformity to vehicle registration formats and significantly improved overall recognition accuracy of the system.
  - Led and organized the annotation of vehicle license plate images, the first dataset of its kind in Bhutan.

The ANPR system is fully operational and deployed on highways in Bhutan [View RBP Press Release ↗].

TECHNICAL SKILLS

Programming language and tools: C, Python, SQL, Linux & Bash, Git, LaTeX

Machine learning frameworks & libraries: PyTorch, TensorFlow, scikit-learn, NLTK, OpenCV, Hugging Face Transformers

Data: NumPy, SciPy, Pandas, MNE-Python, Matplotlib and Seaborn, data visualization, data cleaning, data processing and analysis, feature engineering